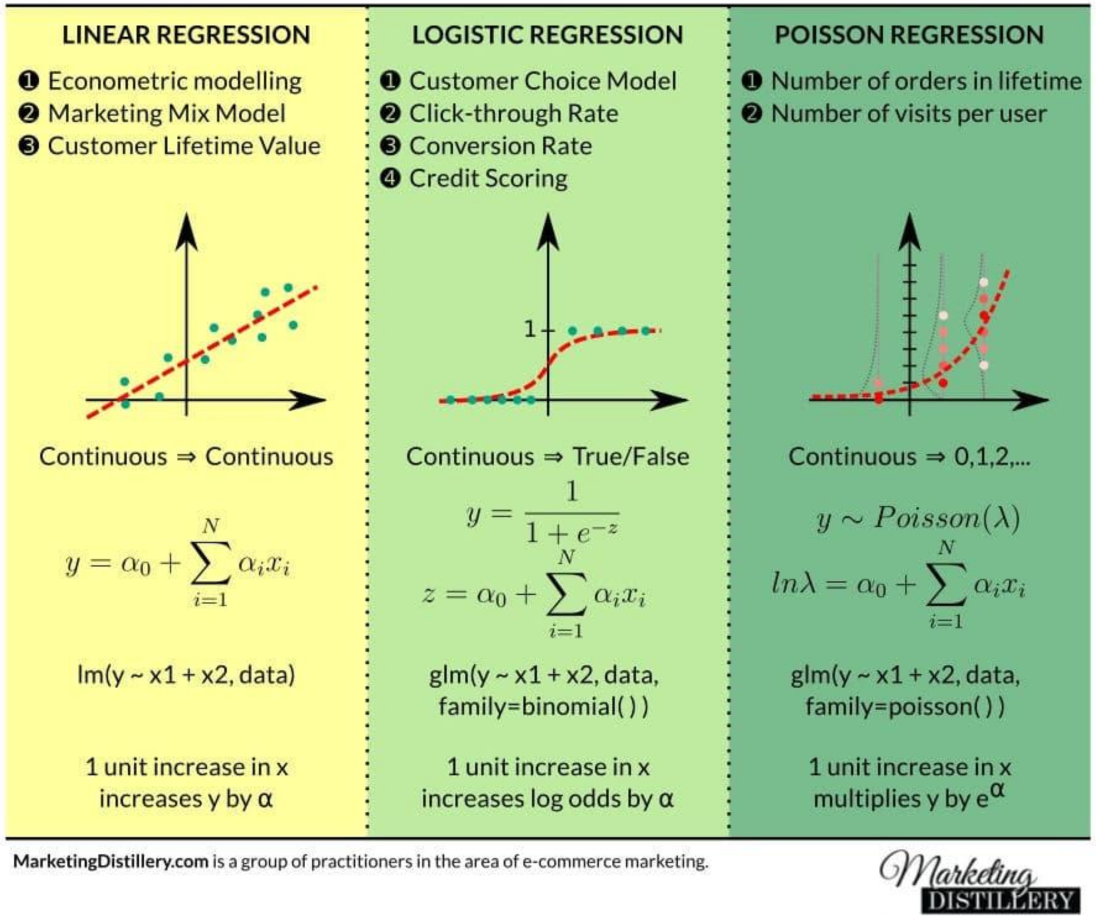
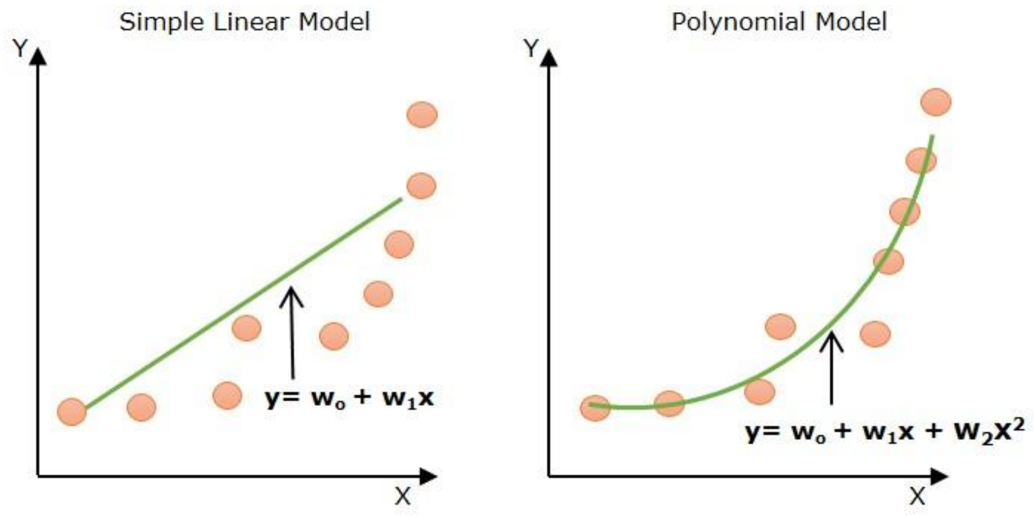


Advanced Regression Concepts

Bending the Rules and Taming the Algorithm

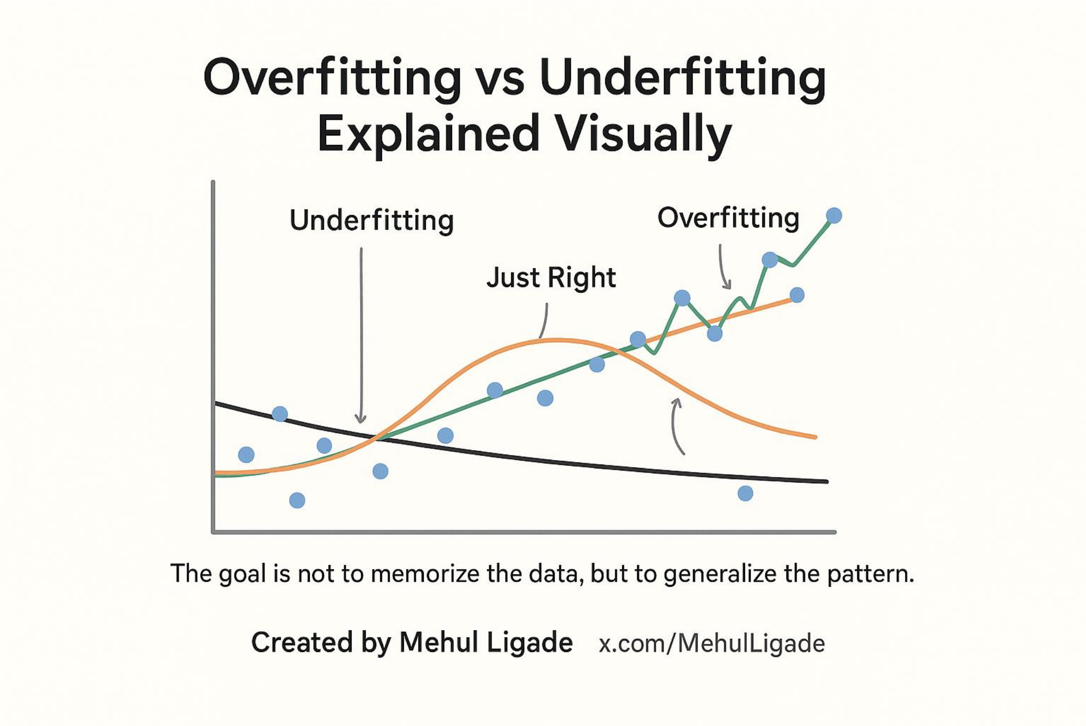
Instructor: Zaryab Ahmad Khan

Polynomial Regression (Bending the Line)



- **The Problem:** Standard Linear Regression assumes data moves in a straight, predictable line. Real-world data (like viral spread, compound interest, or engine heat) often curves.
- **The Solution:** Polynomial Regression. We add "curves" to our math equation.
- **Concept:** Instead of just predicting based on X, we allow the model to learn from X^2 or X^3, letting the line of best fit physically bend to follow the data points.

Overfitting vs. Underfitting (The Goldilocks Problem)



- **Underfitting (Too Simple):** The model is a straight line trying to fit a curved dataset. It learns nothing useful and performs poorly on both training and testing data.
- **Overfitting (Too Complex):** The model connects every single dot perfectly. It has "memorized" the training data but will fail completely when given new, unseen data.
- **The Sweet Spot:** A model that captures the general trend without stressing over every tiny outlier.

Regularization (Ridge & Lasso)

- **The Concept:** Algorithms are greedy; they naturally want to overfit to get a "perfect" score. Regularization is a mathematical technique that gently "punishes" the model for getting too complex.
- **Ridge Regression (L2):** Shrinks the importance of less useful columns but keeps them in the equation.
- **Lasso Regression (L1):** The aggressive approach. It forces the importance of useless columns completely to zero, effectively deleting them and keeping the model simple.